[redacted]

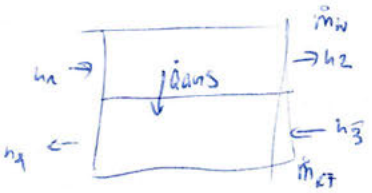
① a) weiter

$$\dot{Q}_{aus} = \dot{m}(h_1 - h_2) + \dot{Q}_R$$

$$\stackrel{!}{=} \underbrace{0.3(-125.67)\frac{kg}{s}\frac{kJ}{kg}}_{\text{aufheizen Strom ein}} + 100\,\frac{kJ}{s} \Rightarrow \boxed{\dot{Q}_{aus} = 62.29\,kW}$$

1b) $$\bar{T}_{KF} = \frac{\int_3^4 T\,ds}{s_3 - s_4} \overset{q_{34}}{=} \frac{\cancel{c_p}(T_4 - T_3)}{\cancel{c_p}\ln\left(\frac{T_4}{T_3}\right)} = \frac{298.15 - 288.15}{\ln\left(\frac{298.15}{288.15}\right)} \Rightarrow 293.12 = \bar{T}_{KF}$$

1c)



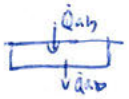
$$-\dot{S}_{erz} = \dot{m}_W(s_2 - s_1) + \dot{m}_{KF}(s_4 - s_3) + \frac{\dot{Q}_{ab}}{\bar{T}_{KF}}$$

$\dot{Q}_{ab}$ ← aus (a), $\bar{T}_{KF}$ ← aus B

$$(s_2 - s_1) = s^\circ(T_2) - s^\circ(T_1) - \cancel{R\ln\left(\frac{p_2}{p_1}\right)} \to 0$$

$$(s_4 - s_3) = s_4^{if}(T_4) - s^{if}(T_3)$$

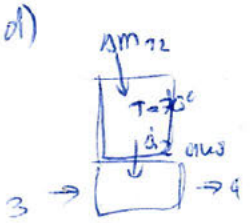
$$\stackrel{!}{=} c^{if}\cdot\ln\left(\frac{T_4}{T_3}\right)$$



$$\dot{S}_{erz} = \dot{Q}_{ab}\cdot\frac{T_R - \bar{T}_{KW}}{T_R\cdot\bar{T}_{KW}} = 62.29\,kW\cdot\frac{373.14 - 293.12}{373.14\cdot 293.12} = 48.91\,\frac{W}{K}$$

---

d)



Energiebilanz in Reaktor / Halboffenes System

$$\Delta E = \Delta m_{12}(h_{in}) + \dot{Q}_{zaus_{12}} \cancel{\dot{W}} + m_{tot}(h_{inside1}) - m_{tot}(h_{inside2})$$

$h_{inside1}$ ← 100°, $h_{inside2}$ ← 70°

$$h_{in} = h(20°C) \overset{A2}{=} 83.96\ kJ/kg$$

$h_{inside} = 130.32\ kJ/kg$ (aus (a), 100°C) (inkl. x_2)

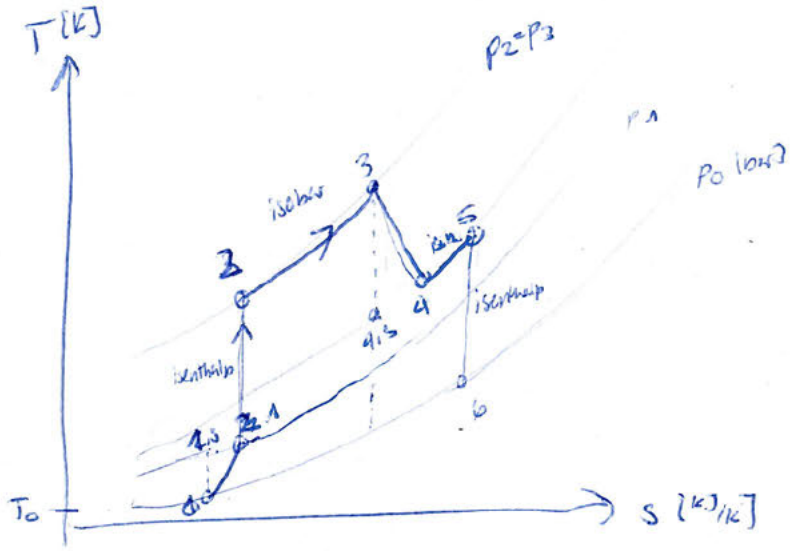
$$\Delta m_{12} = \left(-\dot{Q}_{aus} - m_{tot}(h_{inside1} - h_{inside2})\right)/h_{in}$$

$$\stackrel{!}{=} \left(-35000\,\frac{kJ}{s} + (5755\,kg\cdot(130.32 - 309.64))\right)/83.96 = 8175.8\ kg$$

[redacted]

② Exergie am Flugzeug Triebwerk

a) T-s



$c_p = 1.006$

$\kappa = 1.4$

	T [K]	p [bar]
0		
1	243.10	0.191
↓ isentrop $n=1$		
2		
↓ isentrop $n=1$		
↓ isobar		
3		
↓ $n=0$		
4		
↓		
5	931.9	0.5
↓		
6	319.98	0.191

[redacted]

plug into Energiebilanz gestrmt

$$2\dot{m}(h_6-h_0) = -w_6^2 + w_{luft}^2 \Rightarrow w_{luft}^2 - 2\dot{m}(h_6-h_0) = w_6^2$$

$$\sqrt{200^2 \frac{m^2}{s^2} + 2\cdot 5.8\cdot 71.9} = \boxed{w_6 = 202.07\ m/s}$$

$\dot{m}$ might be wrong
$\Rightarrow$ way too small!

c) $\dot{m}$

$$\Delta e_{xstr} = e_{xstr6} - e_{xstr0}$$

Exergiebilanz

$$\dot{E}_{xstr} = \dot{m}\left(h_6 - h_0 - T_0(s_6-s_0) + \frac{w_{luft}^2 - w_6^2}{2}\right)$$

$$e_{xstr} = h_6 - h_0 - T_0(s_6-s_0) + \frac{w_{luft}^2 - w_6^2}{2}$$

$$e_{xstr} = 71.9 - 293.15\overbrace{(s_6-s_0)}^{0.362} + \left(\frac{200^2 - 510^2}{2}\right)$$

$$\doteq 968656\ kJ/kg$$

$$s_6 - s_0 \overset{ig}{=} s^0(T_6) - s^0(T_0) - \cancel{R\ln\left(\frac{p_6}{p_0}\right)} \to 0$$

$$= c_p \cdot \ln\left(\frac{T_6}{T_0}\right) - \cancel{R\ln\left(\frac{p_0}{p_0}\right)} \to 0$$

$$\frac{c_p}{c_v} = n \Rightarrow c_p = n\cdot c_v = 1.4\cdot 1.006 = 1.4084$$

$$= 1.4084 \cdot \ln\left(\frac{314.45}{293.1}\right) = \boxed{-0.362\ kJ/kgK} = s_6 - s_0$$

d) $e_{xverl} = T_0 \cdot \dot{S}_{erz}$

$$\dot{S}_{erz} = \dot{S}_{verved.} + \dot{S}_{Brennkammer} + \dot{S}_{Turbine}$$

$$= \dot{m}_{ges}(s_1 - s_0) + \dot{m}_K(s_4 - s_3) + \frac{\dot{Q}_B}{T_B}$$

$$\dot{S}_{erz} = (s_1 - s_0) + 1/6.293\,(s_4 - s_3) + \frac{\dot{q}}{T_B}$$

$$\dot{m}_{ges} = \dot{m}_K + \dot{m}_H$$

$$\dot{m}_{ges} = \dot{m}_K + 5.293\dot{m}_K$$

$$\boxed{\dot{m}_K = \dot{m}_{ges}/(6.293)}$$

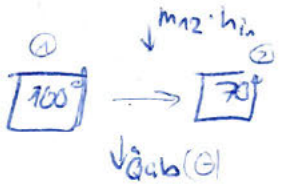
$$(s_1 - s_0) = c_p \cdot \ln\left(\frac{T_1}{T_0}\right) - R\ln\left(\frac{p_1}{p_0}\right)$$

$$(s_4 - s_3) = c_p \ln\left(\frac{T_4}{T_3}\right) - R\ln\left(\frac{p_4}{p_3}\right)$$

[redacted]

1e) ΔS_{12}

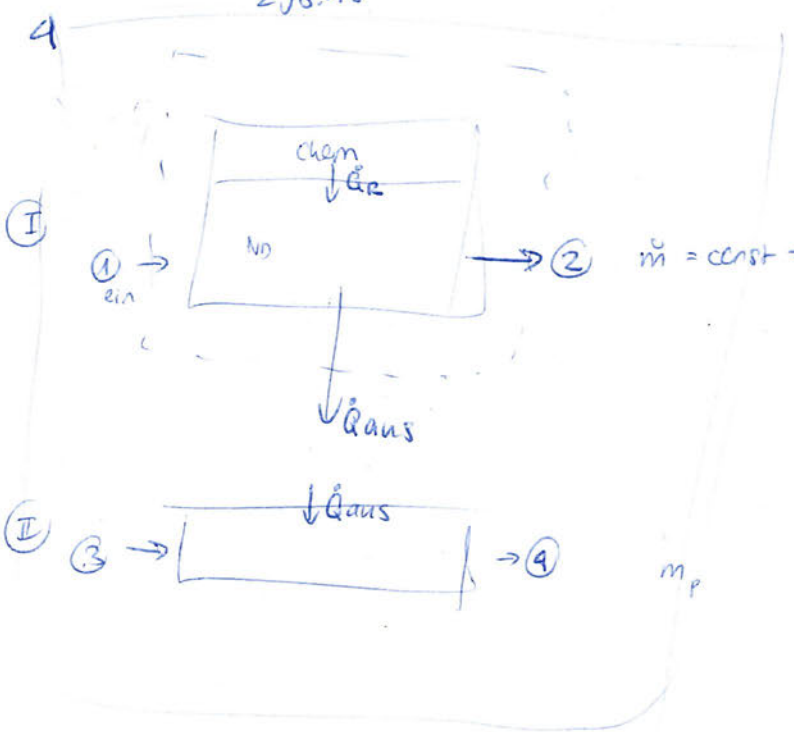
Entropiebilanz



$$\Delta S = \left(m_{tot} + m_{12}\right) s_2 \,(\text{at } T = 70°) - m_1 s_1 \,(\text{at } T = 100) + \frac{\dot{Q}_{ab}}{T_{KW}}$$

~~Q~~ 1

	T [K]	p [bar]	x
1	343.14		0.005
2	373.14		0.005
3	288.15	p_B	
4	298.15	p_B	



$\dot{m} = const = 0.3\ kg/s$ $\quad m_{ges} = 577.5$ (stationär)

a) $\dot{Q}_{aus}$ mit Energiebilanz (I)

$$\boxed{0 = \dot{m}_{geg}(h_1 - h_2) + \dot{Q}_R - \cancel{\dot{W}_T}^{\,0} - \dot{Q}_{aus}}$$

$$(h_1 - h_2)^{Wasser} = h_1(T = \overset{20°}{343.14}) - h_2(T = \overset{100°}{373.14})$$

Dampfanteil $x_1 = x_2$

$h_1 = h_{f1} + x_1(h_{g1} - h_{f1})$ hat T_1

$h_1 = 304.649$

$h_2 = h_{f2} + x_2(h_{g2} - h_{f2})$

$h_2 = 430.32$

$T_{f1} = 292.98$ $\quad h_{f2} = 419.04$ $\quad = x_1 = x_2$

$h_{g1} = 2622.68$ $\quad h_{g2} = 2676.1$ $\quad \boxed{m} \rightarrow \dot{m}$

kJ/kg

⇒ Seite 3

[redacted]

3

b) w_6^2 with $5 \to 6$

$$0 = \dot{m}(h_6 - h_5) + \cancel{\dot{Q}_{ab}}^{\,0} + \dot{m}\left(\frac{w_5^2 - w_6^2}{2}\right)$$

$$2(h_6 - h_5) = w_5^2 - w_6^2 \quad \Rightarrow \quad w_6^2 = w_5^2 - 2(h_6 - h_5)$$

$$(h_6 - h_5) = c_p(T_6 - T_5) = 1.006 \cdot (319.4 - 431.9) = -118.25$$

$$w_6 = \sqrt{220^2 - 2 \cdot -118.25} = 220.26 \text{ m/s}$$

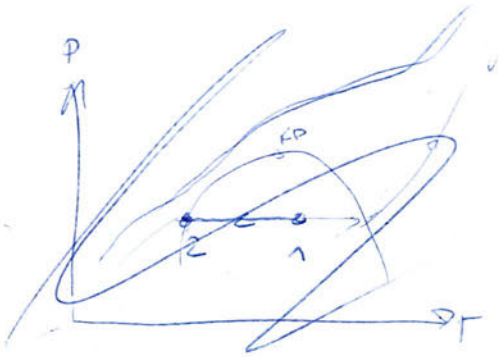
[redacted]

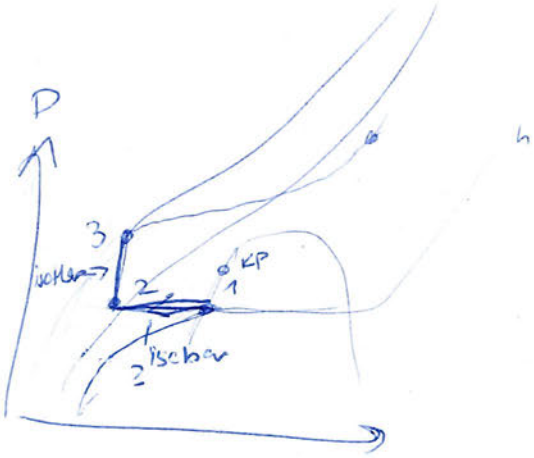
d) $\varepsilon_K = \frac{\dot{Q}_{zu}}{\dot{W}_T} = \frac{|\dot{Q}_K|}{|\dot{Q}_{ab}| + |\dot{W}_K|}$

e) die Temperatur würde die ~~[illegible]~~ bei der gleichen Temp wie das Kühlmedium im Gleichgewicht sein. ($T^* = T_i - 6$)

Kalt kann wegen des 2 HS nicht ~~[illegible]~~ zu warm übertragen werden.

3 ETS





b) $\dot{m}$ durch ~~Energiebilanz~~ um ~~gesamtsystem~~ Verdichter

$\frac{\dot{W}_K}{\dot{m}} = w_K^{rev}$ $\qquad w_K^{rev} = \int_1^2 v\,dp \qquad \Rightarrow \dot{m} = \frac{\dot{W}_K}{w_K^{rev}}$

~~Temp Unterschied:~~ $\frac{T_3}{T_2} = \left(\frac{p_3}{p_2}\right)$

~~Wärmeübertrag~~ $\Delta U = \eta q - W$

c) x_1 direkt nach drossel $\qquad T_{verd} - T_1 - 8K =$

$\boxed{T_1 = T_i - 6K}$ $\qquad$ 4→1 adiabat

$x = \frac{s_1 - s_{1f}}{s_g - s_{1f}}$

$\frac{\dot{U}}{dt} = \dot{Q}\,\frac{T_H - T_K}{T_H \cdot T_K} = 50\,\text{kW}$

$\frac{1}{1 - \frac{T_K}{T_H}}$ $\qquad \frac{1}{1 - \frac{T_K}{T_H}}$

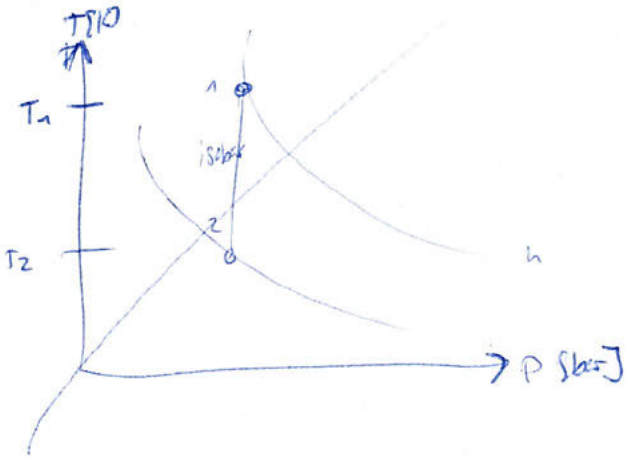
[redacted]

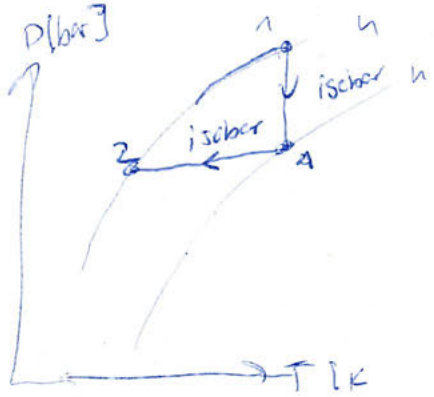
~~③ ETS~~

~~a) p_{01}~~

④ TAB A-10

a)





~~b) $\dot{m}_R$ mit Energiebilanz um~~

	T [K]	p
1 NP ~~vap sat~~		
2 vap sat	Ausfab A10 bei sat	$p_2 = p_1$
3 Vsuperh.	TAB A15 bei 8bar,	8
4 liq sat	31.33	8

[redacted]

6) w_6

Energiebilanz im Gesamtsystem

$w_0 = w_{Luft} = 200 \frac{m}{s}$

$$0 = \dot{m}\left(h_0 - h_6 + \left(\frac{(w_{Luft}^2 - w_6^2)}{2}\right) + \cancel{\dot{Q}}^{0} - \cancel{W}^{0}\right)$$

$$h_0 - h_6 \overset{ig}{=} h(T = 293.14) - h(T = T_6)$$

$$\overset{!}{=} c_p(T_6 - T_0) = 1.006 \cdot (319.4 - 293.14) = \boxed{71.9 \frac{kJ}{kg} = (h_6 - h_0)}$$

$\Rightarrow$ find T_6 via Schubdüse $5 \to 6$

$$\left(\frac{T_6}{T_5}\right) = \left(\frac{p_6}{p_5}\right)^{\frac{n-1}{n}} \Rightarrow T_6 = T_5 \cdot \left(\frac{p_6}{p_5}\right)^{\frac{\kappa-1}{n}}$$

$$\overset{!}{=} 431.0 \cdot \left(\frac{0.191}{0.5}\right)^{\frac{0.4}{1.4}} = 319.4\ K = T_6$$

$\dot{m}_{ges}$ finden mit $2 \to 3$, $\dot{m}_K$

$$\frac{\dot{m}_H}{\dot{m}_K} = 5.293$$

Energiebilanz um Brennkammer

$$0 = \dot{m}_K(h_2 - h_3) + q_B$$

$$(h_2 - h_3) = c_p(T_2 - T_3) = c_p(T_B) = 1.006 \cdot 1289\ [kJ/kg\,K]$$

$$\overset{!}{=} 1296.7\ [kJ/kg\,K]$$

$$\dot{m}_K = \frac{q_B}{h_2 - h_3} = \frac{1195}{1296.7} = 0.9215\ kg/s$$

$$\dot{m}_H = 5.293\,\dot{m}_K = 4.8779\ kg/s$$

$$\dot{m}_{tot} = \dot{m}_K + \dot{m}_H = 5.799\ kg/s \approx \boxed{5.8\ kg/s = \dot{m}}$$